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Classifying Sentinel-2 Satellite Images Based on Tree Coverage

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Introduction		Machine Learning Models		Conclusion
As the number of imaging satellites increases, so too does the amount of image data about our planet's surface. Various subsets of this data have been used to model economic and sociological patterns. However, this imagery could prove useful in identifying trends in Earth's ecosystems. For instance, deforestation and tree density, aspects of our planet that contribute greatly to larger issues, such as climate change. Thus, we sought to train machine learning models that utilize a subset of this data to classify whether a given satellite image contains tree coverage.		Random Forest Decision Trees in Ensemble Training Time: 3.6s (Required Parameter Tuning) Convolutional Neural Network Utilizes Convolutions and Pooling Training Time: 1m 27s Residual Neural Network Groups Convolutions in Identity Blocks to Increase Efficiency Training Time: ~10m Using Google Colab GPU Accel		All three of our models performed well with accuracies greater than 90%, however it is clear our Neural Networks excel over the Random Forest. The simpler CNN and the ResNet34 have similar performances. This was expected considering the small size of our input images and thus less utilization of all layers in the ResNet.
Future Implications				
With satellite image datasets constantly increasing, it is entirely possible to make these models more robust using larger, higher resolution images from other sources, such as Satellite image APIs. In doing so, we will be able to track tree coverage and deforestation rates over large areas by aggregating classifications of smaller areas. This could yield useful information about climate change and the future of our planet as a whole.				
				